Optimal Taxation of Indistinguishable Activities*

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Abstract

Optimal tax policy frequently prescribes different tax treatments for activities that are difficult in practice for tax administrators to distinguish, such as expenditures undertaken for business versus personal purposes. We study the optimal design of income tax deductions for such expenditures under the assumption that a common deduction rate must be applied to both forms of the expenditure. The Pareto optimal common deduction turns on the relative compensated spending response for each margin, weighted by the income tax wedge. We also characterize the welfare gain from distinguishing between the two expenditure forms as a harmonic mean of the compensated responses. We illustrate both results with an application to air travel expenses.

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1 Introduction

Optimal tax research has mostly focused on settings that abstract from the real-world challenges of administering income tax systems. However, such considerations can be important in practice, such as when the optimal tax treatment for a transaction turns on something that is difficult for the tax administrator to observe. For example, if a taxpayer incurs an expense to generate income, the optimal tax treatment is generally to allow a deduction; in contrast, if the motivation for the expense is personal consumption, no deduction should optimally be allowed. Yet a tax administrator cannot easily distinguish between these uses. Many expenditures (e.g., a meal, trip, vehicle, or computer) may be purchased for either business or personal purposes, and it can be difficult for outside observers to readily determine which is which.

In this paper, we study how tax policy should account for expenditures whose appropriate treatment differs according to their purpose when that purpose is not readily observable. We begin with the case in which expenditures are either fully personal or fully business, and the government does not attempt to distinguish between the two categories. In the United States, for example, the government denies deductions for most purchases of day-to-day clothing without any attempt to determine whether the expenditure was undertaken for income-earning reasons. To study optimal policy in this setting, we consider an extension of a standard Atkinson–Stiglitz setup in which consumers can purchase a regular consumption good as well as a business good that contributes to income production.

Our first result is to characterize the Pareto optimal design of income tax deductions that do not distinguish between business and personal expenditures. We show that this optimal common deduction balances two competing considerations. On the one hand, a higher deduction rate reduces the tax distortion to spending on the business input, moving its treatment closer to the first-best benchmark of full deductibility. At the same time, a higher deduction rate subsidizes spending on the consumption good, moving its treatment farther from its no-deduction benchmark. We characterize the deduction rate that optimally balances these opposing forces and show that it depends on the relative compensated responsiveness of spending on the two types of expenditures: the more elastic is one form of the expenditure, the more the optimal common deduction will

reflect that expenditure's treatment under the first-best benchmark.[1] In addition, we show that a higher marginal income tax rate pushes the optimal common deduction upward because incomplete deductibility creates a larger distortion to business spending when the return to that spending is taxed more heavily.

We next turn to settings in which it is feasible for the government to condition a transaction's tax treatment on whether it was undertaken for a business or personal purpose. Tax administrators may, for example, rely on objective evidence bearing on a taxpayer's subjective motivation, such as the broader economic context in which an expenditure occurs or its relationship to the taxpayer's income-producing activities. Rules of this sort are common in practice. Under the U.S. income tax, for example, the deductibility of certain travel expenses turns on the taxpayer's *primary purpose* for undertaking the trip. More generally, many tax rules condition their application on the "facts and circumstances" surrounding a transaction. Although such rules permit tax treatment to be more closely tailored to the nature of an expenditure, they also impose administrative and compliance costs, such as through additional record-keeping and substantiation by taxpayers or through audits, fact-intensive enforcement, and judicial review by the government.

To guide the choice between a pooled deduction rule and a more costly system of differentiated treatment, we characterize the welfare gain from allowing the government to distinguish between business and personal expenditures. We find that the value of differentiation depends on the behavioral responsiveness of both types of expenditures. If personal expenditures are relatively unresponsive to deductibility, the government can set the common deduction rate close to full deductibility, thereby largely eliminating the distortion to business spending at little cost on the personal-spending margin. Conversely, if business expenditures are relatively unresponsive, the common deduction can be set close to zero. The gains from differentiation are therefore largest when both types of expenditures respond substantially to their tax treatment. In that case, any common deduction necessarily creates meaningful distortions on at least one of the two margins. The value of differentiation is also increasing and convex in the marginal income-tax rate: a higher

[1] Applied to our example of day-to-day clothing, this logic implies that a no-deduction rule is optimal precisely when the work-related component of such spending is unresponsive to its tax treatment. A deduction would then subsidize personal consumption without meaningfully encouraging business use.

marginal rate both increases the effect of deductibility on taxpayers' choices and magnifies the fiscal consequences of the resulting behavioral responses. Our formula for the welfare gain from differentiation thus provides a benchmark for assessing when the benefits of more finely targeted tax treatment are large enough to justify the additional administrative and compliance costs it requires.

We then illustrate how these characterizations can be applied, taking the deductibility of air travel expenses as an example. Income tax systems typically restrict deductions to travel undertaken primarily for business, but determining the purpose for which travel was undertaken can be costly. We characterize the common deduction rate that would apply if deductibility was not conditioned on purpose, as well as the welfare gain (gross of any administrative costs) from limiting deductibility to business travel. Published fare elasticities indicate that personal air travel is considerably more fare-responsive than business travel. Because the optimal common rate reflects the relative responsiveness of the two margins, it is low (approximately 5%) even though business travel would be fully deductible under differentiated treatment. We estimate that the welfare gain from limiting deductibility to business air travel is approximately 2.6% of expenditures in that category.

Finally, we extend the analysis in a number of directions. In particular, we allow for cross-price effects between the expenditures in the pooled category and we consider a deduction rate that must be uniform across taxpayers. Moreover, we consider settings with multiple goods that differ in the extent to which they provide consumption benefits or contribute to income production. In this more general setting, the optimal common deduction can be understood as a weighted average of the tax treatment appropriate for the different goods in the category, with weights reflecting both spending responsiveness and exposure to the income-tax wedge.

Our results build on an established literature considering how the design of optimal tax systems should account for the administrative technology through which taxes are implemented; for a comprehensive discussion, see Slemrod and Gillitzer (2014). Some prior work in this literature takes administrability constraints as given and asks how tax rates should adjust (e.g., Stern, 1982; Slemrod and Yitzhaki, 1994; Gordon and Li, 2009). Other work treats aspects of administrability as endogenous and studies the value of greater accuracy or observability in tax administration

(e.g., Kaplow, 1998; Slemrod and Traxler, 2010). Our current results relate to both strands of this literature: we characterize optimal deductibility when distinguishability is taken as given and characterize the welfare value of relaxing this constraint.

The prior work to which we most closely relate is Baake et al. (2004), who also study optimal income tax deductions in light of the difficulty of distinguishing consumption from work-related expenditures. Our results build on theirs in two ways. First, we more fully characterize the design of the optimal deduction: whereas Baake et al. show that the optimal common deduction rate is between 0% and 100%, we provide an analytic formula for the optimal rate. The formula makes transparent the factors that determine the optimal degree of deductibility and how changes in those factors affect the optimal policy. Our second contribution relative to Baake et al. is to characterize the welfare gain from permitting differentiated treatment of business inputs and consumption goods. This welfare gain provides the benefit side of the policy tradeoff over whether to incur the additional administrative and compliance costs required to differentiate between the two types of expenditures.

Also related is prior work on the optimal deduction for mixed-purpose expenditures that both generate income and provide consumption benefits (Richter, 2006; Goldin et al., 2026).[2] In that setting, the two purposes are bundled within a single good whose composition is observable to the tax authority, making it optimal to tailor the deduction rate directly to that composition. In the current framework, by contrast, the purposes are attached to separate expenditures that the tax authority cannot distinguish, requiring a common tax treatment for both. The optimal deduction rate therefore depends on the relative responsiveness of the two margins, since the same rate must balance the distortions associated with each margin.[3] Bastani et al. (2019) and Koehne and Sachs (2022) study deductions and subsidies for work-related goods that complement labor supply, such as child care and household services. That work likewise takes the identity of the work-related good

[2]Outside of economics, legal scholarship considers a range of related topics; see Halperin (1973); Griffith (1993); Givati (2020).

[3]The two settings therefore yield different policy prescriptions. For a mixed-purpose good, Goldin et al. (2026) show that the relevant question is whether marginal spending is primarily income-producing or consumption-oriented; for example, additional spending on a work-related phone may largely go towards consumption features such as a better camera. In the current setting, by contrast, even a purely business expenditure may optimally receive a low common deduction rate if spending on that margin is relatively unresponsive compared to (observationally equivalent) personal spending. See Section 6.3 for additional discussion.

as known, whereas we ask how the deduction should be designed when the tax authority cannot tell which expenditures are work-related.

Finally, our results relate to a broader literature in public economics on the optimal design of imperfectly targeted or second-best policy instruments. For example, Diamond (1973) studies corrective pricing when heterogeneous activities generate different externalities, but the policymaker is constrained to impose a common price. Our optimal deduction rule has a similar structure: business and personal expenditures ideally call for different deduction rates, but when they cannot be distinguished, the government must choose a common rate that balances the distortions on the two margins according to their behavioral responsiveness. An important difference is that our externality is not technological and cross-agent, but a within-taxpayer fiscal spillover: an individual's expenditure choices move their own taxable income and hence government revenue. This is why our weights carry an income-tax wedge that has no counterpart in the corrective-pricing setting. Other recent contributions that study related problems of imperfectly targeted corrective policy with similar themes to our analysis include Knittel and Sandler (2018), Gauthier and Henriet (2023), Dávila and Walther (2026), and Weisbach (2026).

The rest of the paper is organized as follows. Section 2 presents the model. Section 3 characterizes the optimal common deduction rate. Section 4 characterizes the welfare gain from allowing the tax authority to differentiate between business and personal expenditures. Section 5 presents an illustration of our results, applying the theory to the setting of air travel expenses. Section 6 considers several extensions of the baseline model, and Section 7 concludes. The Appendix contains proofs of the main results.

2 Setup

Our analysis starts from the classic Mirrlees (1971) model of optimal income taxation, in which individuals of different skill levels choose their labor supply and use their after-tax labor income to purchase a numeraire consumption good. We extend this model by incorporating an additional "consumption good," which generates utility, as well as a "business good," which contributes to income production. We will initially assume that these two additional goods are observationally

equivalent to the government so that tax policy cannot distinguish between them. Under this constraint, we consider the optimal common deduction rate that should apply to the two goods.

2.1 Environment

The economy is populated by a continuum of individuals who differ in skill $w \in \mathcal{W} := (\underline{w}, \overline{w}) \subset \mathbb{R}_{++}$, distributed according to a smooth density $f : \mathcal{W} \to \mathbb{R}_{++}$ with full support. An individual with skill w who supplies labor effort l earns labor income:

$$x = wl. \tag{1}$$

Preferences are weakly separable:

$$U(c, d, l, w) = u\big(v(c, d), l, w\big), \tag{2}$$

where c denotes the numeraire consumption good and d denotes the additional consumption good that – along with the business good introduced below – will be our focus. The subutility function $v : \mathbb{R}^2_+ \to \mathbb{R}$ is strictly increasing in both arguments and strictly concave; the outer utility function $u : \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_{++} \to \mathbb{R}$ is concave in its first two arguments, strictly increasing in its first argument, and strictly decreasing in its second argument. As in the Atkinson and Stiglitz (1976) framework, the consumption subutility is identical across individuals, while labor preferences may vary with skill.

In addition to consumption goods c and d, individuals may purchase a business good e that contributes to total income. Total income is given by the strictly concave function $y : \mathbb{R}^2_+ \to \mathbb{R}_+$:

$$y = y(x, e), \tag{3}$$

with $y_x > 0$ and $y_e \geq 0$. The sign of the cross-partial derivative $y_{x,e}$ is unrestricted. The economy-wide distribution of labor incomes, chosen as a function of government policy, is denoted by $H(\cdot)$.

The respective prices of d and e are denoted by p_d and p_e. We assume that these prices are

exogenously fixed – for example, because the goods are supplied competitively at constant marginal cost – so that changes in the deduction rate do not affect their market prices. The price of c is normalized to one.

2.2 Tax System with Pooled Deduction

We initially assume that the government cannot distinguish between expenditures on d and e. We therefore study tax systems of the form:

$$T\big(y(x,e) - \alpha(x)(p_d d + p_e e) + \gamma(x)\big), \tag{4}$$

where $T : \mathbb{R} \to \mathbb{R}$ is the non-linear income tax schedule and $\alpha : \mathbb{R}_+ \to \mathbb{R}$ is the common deduction rate that applies to total expenditures on d and e. We assume that this deduction rate α can vary by labor income. The final term in (4), $\gamma : \mathbb{R}_+ \to \mathbb{R}$, is an auxiliary component of the income tax base that can be perturbed as part of the reforms we consider below. Taxable income is therefore given by

$$z \equiv y(x,e) - \alpha\left(x\right)\left(p_d d + p_e e\right) + \gamma\left(x\right). \tag{5}$$

We assume that T, α, and γ are continuously differentiable, with $0 < T'(z) < 1$ and $T''(z)$ defined for all z.

At the baseline policy, let

$$P_d(x) \equiv p_d\big(1 - \alpha(x)T'(z(x))\big), \qquad P_e(x) \equiv p_e\big(1 - \alpha(x)T'(z(x))\big) \tag{6}$$

denote the after-tax prices of d and e, incorporating the allowable deduction.

It will be convenient to present our results using conditional compensated (Hicksian) demand for d and e, denoted by d^c and e^c, with respect to changes in P_d and P_e. These compensated demands are conditional in the sense that they are defined from the expenditure-minimization problem in which consumption subutility is held constant at each level of labor income (x), rather than only at the taxpayer's initial optimized choice.[4]

[4]In empirical demand analysis, compensated elasticities are often recovered from estimated uncompensated de-

Finally, for ease of exposition, we assume that cross-price substitution effects between d and e are zero:

$$\frac{\partial d^c}{\partial P_e} = \frac{\partial e^c}{\partial P_d} = 0. \tag{7}$$

We relax this assumption in Section 6.1.

2.3 Individual Problem

The individual optimization problem is given by:

$$\max_{c,d,e,l} u\left(v\left(c,d\right),l,w\right) \quad \text{s.t.} \quad c + p_d d + p_e e = y\left(x,e\right) - T\left(y\left(x,e\right) - \alpha\left(x\right)\left(p_d d + p_e e\right) + \gamma\left(x\right)\right). \tag{8}$$

Because the utility function is weakly separable, we can simplify the analysis by decomposing the problem into two stages. Given labor income $x = wl$, the individual chooses (c, d, e) to maximize the subutility function:

$$v^*\left(x; T, \alpha(x), \gamma(x)\right) = \max_{c,d,e} v(c,d) \quad \text{s.t.} \quad c + p_d d + p_e e = y(x,e) - T\big(y(x,e) - \alpha(x)(p_d d + p_e e) + \gamma(x)\big). \tag{9}$$

Note that e does not enter the utility function directly; it affects utility only through its impact on the budget constraint.

Anticipating the v^* function, the individual chooses labor effort to maximize:

$$U^*(w; T, \alpha, \gamma) = \max_{l} u\big(v^*\left(wl; T, \alpha(wl), \gamma(wl)\right), l, w\big). \tag{10}$$

At an interior solution, the first-order conditions for d and e from (9) are:

$$v_d = \big(1 - \alpha T'\big) p_d v_c, \tag{11}$$

$$p_e(1 - \alpha T') = y_e(1 - T'). \tag{12}$$

mand responses by applying a Slutsky adjustment using the associated income or expenditure effect. This procedure identifies the elasticities defined here when the demand system conditions on labor income (x) and estimates substitution responses within the fixed-x consumption problem, for example by combining uncompensated price coefficients with income-effect estimates from a demand system conditional on total expenditure or disposable resources.

The first condition (11) states that the marginal rate of substitution between the consumption good d and the numeraire c equals the after-tax price of d accounting for deductibility. The second condition (12) equates the marginal after-tax cost of investing in the business good e to its marginal after-tax return.

It will be useful to express our results in terms of the conditional compensated own-price elasticities of d and e. Define

$$\varepsilon_d^c(x) \equiv -\frac{\partial d^c(x)}{\partial P_d}\frac{P_d}{d(x)}, \qquad \varepsilon_e^c(x) \equiv -\frac{\partial e^c(x)}{\partial P_e}\frac{P_e}{e(x)}, \tag{13}$$

where P_d and P_e denote the respective after-tax prices defined above.

Finally, to simplify notation, let $q_d(x) \equiv d(x)$ and $q_e(x) \equiv e(x)$, and define

$$S_i(x) \equiv \varepsilon_i^c(x)\, p_i\, q_i(x), \qquad i \in \{d, e\}. \tag{14}$$

The term $S_i(x)$ captures the compensated spending response to a change in the after-tax price of good i. It is larger when expenditure on the good is greater or when demand is more price-responsive. Since the common deduction rate changes the after-tax prices of both goods proportionally, the relative sizes of $S_d(x)$ and $S_e(x)$ determine which behavioral margin responds more strongly to the deduction.

3 Optimal Common Deduction Rate

We now characterize the Pareto optimal common deduction rate using a tax-reform argument in the spirit of Kaplow (2006). Specifically, we consider a small perturbation to the deduction rate at a particular level x of labor income, accompanied by an adjustment to $\gamma(x)$ that is chosen to leave individual subutility unchanged. If such a subutility-preserving reform generates an increase in tax revenue, the additional revenue can be distributed to taxpayers lump-sum to yield a Pareto improvement.

We therefore characterize the Pareto optimal deduction rate by determining the deduction rate

schedule for which no such compensated reform generates an increase in revenue. This approach yields the following necessary condition for Pareto optimality. A formal proof is provided in Appendix A.

Proposition 1 (Optimal Common Deduction Rate)**.** *Consider tax functions of the form*

$$T\left(y\left(x,e\right)-\alpha\left(x\right)\left(p_{d}d+p_{e}e\right)+\gamma\left(x\right)\right),$$

where $\alpha(x)$ denotes the common deduction rate for expenditures on d and e. Absent cross-price substitution effects for d and e, any interior Pareto optimal deduction rate at labor income level x satisfies

$$\alpha(x)=\frac{S_{e}(x)}{S_{e}(x)+\left(1-T'(z(x))\right)S_{d}(x)}$$

where $S_i(x)=\varepsilon_i^c(x)\,p_i\,q_i(x)$ for $i\in\{d,e\}$.

To understand the intuition for Proposition 1, it is useful to start from the fully differentiated benchmark. If the government could distinguish the two goods, it would allow taxpayers to fully deduct expenditures on the business good e, thereby removing expenditures on that good from the income tax base (c.f., Diamond and Mirrlees, 1971). By contrast, it would not allow deductions for expenditures on the consumption good d, since d directly enters utility and is not an input into income production (c.f., Atkinson and Stiglitz, 1976). Pooling creates a conflict between these two prescriptions: increasing the common deduction rate $\alpha(x)$ moves the tax treatment of e closer to its ideal rule but simultaneously moves the tax treatment of d farther away from its ideal rule.

The optimal common deduction reduces to the differentiated benchmark in the limiting case that expenditures on only one of the goods is responsive to changes in its after-tax price. If spending on d is unresponsive, so that $\varepsilon_d^c(x)=0$, then the consumption good distortion is irrelevant at the margin and the optimal common rate is $\alpha(x)=1$. In that case, pooling does not create a cost on the d margin, so the government can set the deduction rate as if only the business good existed. Conversely, if spending on e is unresponsive, so that $\varepsilon_e^c(x)=0$, then deductibility no longer serves a useful incentive role for the business good, and the optimal common rate is $\alpha(x)=0$. In that case, the government can set the deduction rate as if only the consumption good existed.

Between these two extremes, the optimal policy is a partial deduction. The reason is that a higher common deduction rate creates two offsetting fiscal externalities. When taxpayers purchase an additional unit of e, taxable income changes by $y_e(x, e(x)) - p_e\alpha(x)$: the input raises pre-tax income by $y_e(x, e(x))$, but also increases deductions by $p_e\alpha(x)$. The resulting fiscal gain is therefore

$$T'\left(z(x)\right)\big(y_e(x, e(x)) - p_e\alpha(x)\big).$$

By contrast, when taxpayers purchase an additional unit of d, pre-tax income does not rise, but deductions increase by $p_d\alpha(x)$. The resulting fiscal cost is therefore

$$T'\left(z(x)\right)p_d\alpha(x).$$

The compensated reform used to derive Proposition 1 removes the mechanical effect of changing the deduction rate, leaving just these behavioral fiscal externalities. Thus, at the optimum, a marginal increase in the common deduction rate must have no first-order effect on revenue:

$$T'\left(z(x)\right)\Big[\big(y_e(x, e(x)) - p_e\alpha(x)\big)e_\alpha(x) - p_d\alpha(x)d_\alpha(x)\Big] = 0 \qquad \text{for all } x \tag{15}$$

where $d_\alpha(x)$ and $e_\alpha(x)$ are the derivatives of spending on d and e with respect to a change in α (formally defined in Appendix A). This condition says that the fiscal gain from the induced increase in spending on the business good must exactly offset the fiscal cost from the induced increase in spending on the consumption good. If the e response is large relative to the d response, the common deduction rate should be high; if the d response is large relative to the e response, the common deduction rate should be low.

One final way to view Proposition 1 is as a fiscal-externality analogue of Diamond's (1973) imperfect corrective pricing result. In Diamond's setting, a single price must apply to activities that generate different externalities, so the optimal common price averages the differentiated corrective prices, with greater weight on more price-responsive activities. Here, full deductibility for the business good and zero deductibility for the consumption good are the two differentiated bench-

marks. Because the government cannot distinguish the goods, the optimal common deduction lies between them, with greater weight placed on the benchmark associated with the more responsive expenditure margin (and with the weights adjusted for the income-tax wedge).

The factor $1-T'(z(x))$ in the denominator in Proposition 1 reflects the fact that the individual benefit of additional expenditures on e operates through the income tax base. Because the return to e is taxed, taxpayers capture only the after-tax return to the input. The taxpayer's first-order condition for e implies

$$y_e(x, e(x)) - p_e\alpha(x) = \frac{p_e(1-\alpha(x))}{1-T'(z(x))}.$$

Thus, a gap between the common deduction rate and full deductibility, $1-\alpha(x)$, is magnified by the factor $\frac{p_e}{1-T'(z(x))}$ on the input margin.

4 Welfare Gain from Differentiation

We now ask how much the tax authority's inability to distinguish the consumption good d from the income-generating input e costs in welfare terms. The answer matters because observability is itself partly endogenous: by devoting auditing and documentation resources to a category of spending, the authority can often separate consumption from input use that would otherwise be pooled. Since such monitoring is costly, the welfare gain from differentiation identifies the categories where this investment is most worthwhile.

To characterize this gain, we compare the common deduction rate to a fully differentiated benchmark. To define the differentiated benchmark, suppose taxable income can instead take the form

$$z(x) = y(x, e(x)) - \alpha_d(x)p_d d(x) - \alpha_e(x)p_e e(x) + \gamma(x), \tag{16}$$

where the government may set separate deduction rates for the two goods. The differentiated optimum sets

$$\alpha_d(x) = 0, \qquad \alpha_e(x) = 1,$$

so that consumption expenditures are not deductible and expenditures on the income-generating

input are fully deductible.

We approximate the gain from moving from the optimal pooled policy to this differentiated benchmark using a Harberger-style local approximation. In particular, we evaluate the marginal tax rate and compensated own-policy response slopes at the pooled optimum and treat them as locally constant as the two deduction rates are moved to their differentiated values. The proof in Appendix B then expresses those baseline slopes using the Hicksian elasticities from Proposition 1. As in that proposition, the approximation is pointwise in labor income. Then, the total gain is obtained by aggregating across the income distribution.

Proposition 2 (Welfare Gain from Differentiation)**.** *Consider tax functions of the form*

$$T\left(y\left(x,e\right)-\alpha\left(x\right)\left(p_{d}d+p_{e}e\right)+\gamma\left(x\right)\right),$$

where $\alpha(x)$ is the optimal common deduction rate characterized in Proposition 1. Absent cross-price substitution effects for d and e, the welfare gain (in units of fiscal revenue) from replacing the common deduction with the differentiated rates $\alpha_d(x) = 0$ and $\alpha_e(x) = 1$ is approximately given by

$$\Delta W \approx \frac{1}{2}\int \frac{\left(T'(z(x))\right)^{2}}{1-T'(z(x))}\frac{S_{e}(x)S_{d}(x)}{S_{e}(x)+S_{d}(x)}\,\mathrm{d}H(x), \tag{17}$$

where $S_d(x)$ and $S_e(x)$ are the compensated spending-response terms defined in (14)*.*

To understand Proposition 2, first consider the two distortions created by pooling. Because the common deduction rate is below one, expenditures on the business good remain partially exposed to the income tax, discouraging the use of e. At the same time, because the common rate is above zero, expenditures on the consumption good receive a tax subsidy, encouraging the use of d. Differentiation eliminates both distortions by raising the deduction for e from $\alpha(x)$ to one and lowering the deduction for d from $\alpha(x)$ to zero.

Under the Harberger approximation, the local welfare gain at income level x is the sum of the two corresponding distortion triangles. The first is the gain from eliminating the remaining distortion on the business input, while the second is the gain from eliminating the subsidy to consumption. In each case, the size of the triangle is increasing in the compensated spending response on the

relevant margin and in the square of the distance between the common deduction rate and that good's differentiated benchmark.

The common deduction characterized in Proposition 1 is chosen precisely to balance the marginal costs of these two distortions. Substituting that optimal rate into the two-triangle expression collapses the behavioral component of the welfare gain to

$$\frac{S_e(x)S_d(x)}{S_e(x) + S_d(x)},$$

which is proportional to the harmonic mean of the two compensated spending responses. This structure captures the central economic implication of the result: differentiation is valuable only to the extent that both margins are behaviorally important. If either $S_e(x)$ or $S_d(x)$ is zero, the common deduction rate can be set at the differentiated benchmark for the margin that responds, and differentiation yields no welfare gain under the approximation. When both margins respond substantially, by contrast, any common deduction necessarily leaves a meaningful distortion on at least one of them.

The harmonic-mean structure also makes clear when pooling is most costly. Holding $S_e(x) + S_d(x)$ fixed, the welfare gain from differentiation is largest when $S_e(x) = S_d(x)$. Thus, the greatest gains arise when the two uses generate similarly important compensated spending responses but call for opposite tax treatment. When one response is much larger than the other, the optimal common deduction rate can be set close to the differentiated treatment appropriate for the more responsive margin, leaving relatively little welfare to be gained from distinguishing the goods.

The remaining factor,

$$\frac{\big(T'(z(x))\big)^2}{1 - T'(z(x))},$$

captures the role of the income-tax wedge. One factor of $T'(z(x))$ reflects that a change in deductibility changes the goods' after-tax prices in proportion to the marginal tax rate, while the second reflects the revenue consequence of the resulting behavioral response. The factor $1/\big(1 - T'(z(x))\big)$ reflects the business-input margin: because taxpayers retain only the after-tax return to an income-generating expenditure, incomplete deductibility creates a larger distortion

to that margin when the marginal tax rate is higher. Taken together, these three effects make the welfare cost of pooling rise more than proportionally with the marginal tax rate, holding the compensated spending responses fixed.

In applications, Proposition 2 therefore provides a benchmark for deciding where administrative differentiation is most worthwhile. The largest potential gains arise for categories in which both business and personal expenditures respond substantially to their tax treatment and where the relevant marginal income-tax rates are high. When one component is relatively unresponsive, the common deduction can instead be tailored to the responsive margin at relatively low efficiency cost.

Finally, using $S_e S_d/(S_e + S_d) \leq \frac{1}{4}(S_e + S_d)$, with equality if and only if $S_e = S_d$,[5] Proposition 2 implies that ΔW is approximately bounded by

$$\frac{1}{8} \int \frac{\big(T'(z(x))\big)^2}{1 - T'(z(x))} \Big(S_e(x) + S_d(x) \Big) \, \mathrm{d}H(x). \tag{18}$$

This bound is useful because it involves only the marginal tax rate and the *sum* of the two compensated spending responses. Because the deduction rate moves the after-tax prices of d and e in the same proportion, the response of total claimed expenditure to that rate recovers $S_e(x) + S_d(x)$ without requiring to separate the two goods. A tax authority can therefore bound the value of distinguishing the two expenditure types using only variation generated by a policy that does not distinguish them.

5 Illustration: Air Travel Expenses

To illustrate our results, we consider the deductibility of air travel expenses. Because expenditures on air travel may be undertaken for either business reasons – e.g., meeting with a client – or personal reasons – e.g., a relaxing holiday – deductibility would ideally turn on the taxpayer's purpose for undertaking the expense. In practice, however, it can be burdensome for both taxpayers and the tax authority to establish the purpose for which travel is undertaken, which requires identifying supporting evidence (e.g., receipts, itineraries, communications) and using that evidence to assess

[5] Note that $(S_d - S_e)^2 \geq 0$ can be rearranged as $(S_e + S_d)^2 \geq 4 S_e S_d$, and dividing by $4(S_e + S_d) > 0$ establishes the inequality used in the text.

the convincingness of the taxpayer's claimed business rationale. Notwithstanding that burden, current law in many if not all income tax systems limits deductibility to business air travel. In the United States, for example, deductibility is limited to travel expenses for which the taxpayer's primary purpose in undertaking the travel is conducting business. In this section, we apply our results to characterize the optimal common deduction for air travel expenses when both business and personal travel can be deducted, as well as to quantify the welfare gain from limiting deductibility to air travel that is undertaken for a business purpose.

We begin with the optimal common deduction rate that would apply under pooling. Let E denote expenditure on business airfare and D expenditure on personal airfare, so that the two compensated spending responses are $S_e = \varepsilon_e^c E$ and $S_d = \varepsilon_d^c D$. Writing $s_e \equiv E/(E+D)$ for the business share of total airfare expenditure, Proposition 1 gives the optimal common deduction rate under pooling:

$$\alpha = \frac{\varepsilon_e^c s_e}{\varepsilon_e^c s_e + \big(1 - T'\big)\varepsilon_d^c \big(1 - s_e\big)}. \tag{19}$$

For purposes of this stylized illustration, we abstract from heterogeneity in budget shares, elasticities, and marginal tax rates across taxpayers.

A consistent empirical finding is that business air travel is substantially less price-elastic than personal travel (InterVISTAS Consulting Inc., 2007). For example, the UK Department for Transport reports long-run fare elasticities of approximately 0.2 for business travel and 1.1 for leisure travel (UK Department for Transport, 2022). Guided by these estimates, we set $\varepsilon_e^c = 0.2$ and $\varepsilon_d^c = 1.1$.

The business-expenditure share s_e that enters (19) is the share that would prevail under the counterfactual pooled regime and is therefore not directly observed. We map the observed expenditure composition under differentiated treatment into this counterfactual using the same elasticities. Let E_0 and D_0 denote business and personal airfare expenditures under the current differentiated regime, in which business airfare is fully deductible and personal airfare is not deductible. The UK International Passenger Survey indicates that business trips accounted for approximately 10 percent of outbound air trips by UK residents in 2024 but, because business travelers tend to pay higher fares, approximately 16 percent of total airfare expenditure (Office for National Statistics,

2025). We therefore take

$$s_{e,0} \equiv \frac{E_0}{E_0 + D_0} = 0.16$$

as our baseline and normalize $E_0 + D_0 = 1$, so that $E_0 = 0.16$ and $D_0 = 0.84$.

To estimate the corresponding expenditure share under pooling, we assume constant elasticities for both goods. Under differentiated treatment, the after-tax price of business airfare is proportional to $1 - T'$, while the after-tax price of personal airfare is proportional to one. Under pooling with a common deduction rate α, both after-tax prices are proportional to $1 - \alpha T'$. Holding pre-tax fares fixed, the implied expenditures under pooling are therefore

$$E(\alpha) = E_0 \left(\frac{1 - \alpha T'}{1 - T'} \right)^{-\varepsilon_e^c} \quad \text{and} \quad D(\alpha) = D_0 \left(1 - \alpha T'\right)^{-\varepsilon_d^c}, \tag{20}$$

so that the business-expenditure share under pooling is

$$s_e(\alpha) = \frac{E(\alpha)}{E(\alpha) + D(\alpha)}. \tag{21}$$

The optimal pooled policy is therefore characterized by the solution to a fixed point problem: the deduction rate determines the expenditure composition through (20), while that expenditure composition determines the optimal deduction rate through (19).

For a stylized numerical illustration, set $T' = 0.4$. At the observed differentiated-regime business share of $s_{e,0} = 0.16$, solving (19)–(21) gives $\alpha^* \approx 0.049$ and $s_e(\alpha^*) \approx 0.145$. The optimal common deduction rate is therefore approximately 5 percent. The low deduction rate reflects the large difference in spending responsiveness across the two uses. Business airfare is, by construction of our setting, an expenditure that would be fully deductible if its purpose were observed. Yet when it must be pooled with substantially more responsive personal airfare, the optimal common rate is close to zero.

Next, we apply Proposition 2 to quantify the value of limiting deductibility of air travel expenses to those undertaken for business purposes. Scaling (17) by E at the pooled optimum and using

$S_e = \varepsilon_e^c E$ lets us write

$$\frac{\Delta W}{E} \approx \frac{(T')^2 \varepsilon_e^c}{2(1-T')} \frac{S_d}{S_e + S_d}. \tag{22}$$

The first factor is a ceiling on the gain. It depends only on the marginal tax rate and the business fare elasticity and equals 2.67 percent at $T' = 0.4$ and $\varepsilon_e^c = 0.2$. The second factor is the share of the total compensated spending response that falls on the personal margin. Because personal airfare is much more elastic than business airfare, this share is large in our baseline calculation. At the pooled optimum implied by $s_{e,0} = 0.16$, (22) evaluates to $\Delta W/E \approx 2.59\%$. Thus, the welfare value of distinguishing business from personal airfare is approximately 2.6 percent of business airfare expenditure under the counterfactual pooled regime, or $2.59\%\,(0.145/0.160) \approx 2.35\%$ of such expenditures under the current differentiated regime.

The qualitative conclusions are fairly robust to the assumed business share of airfare expenditure. To illustrate, the following table considers otherwise identical economies with larger business-airfare shares under differentiated treatment. For each value of $s_{e,0}$, we use (20) to calculate the expenditure composition under pooling and then solve for the corresponding optimal deduction and welfare gain:

$s_{e,0}$	0.16	0.30	0.40	0.50
$s_e(\alpha^*)$	0.145	0.272	0.363	0.456
α^*	0.049	0.101	0.147	0.202
$\Delta W/E$ (%)	2.59	2.50	2.42	2.31

As the business share rises, the business margin receives greater weight and the optimal common deduction rate increases. But even when business airfare accounts for half of airfare expenditure under differentiated treatment, the optimal common rate is only about 20 percent because the personal margin remains much more responsive. For the same reason, the welfare gain from differentiation declines only modestly, from approximately 2.6 to 2.3 percent of business airfare expenditure. The two results reflect the same underlying asymmetry in behavioral responses: the relatively elastic personal margin keeps the optimal common rate low, while also causing the harmonic-mean term in Proposition 2 to remain close to the ceiling determined by the less elastic business margin.

We interpret these results as an illustration of the mechanism rather than robust policy pre-

scriptions. First, the constant-elasticity assumption extrapolates the reported fare elasticities over a substantial price change. Second, the published fare elasticities we rely on are long-run demand responses, whereas our theory calls for compensated responses to after-tax prices conditional on labor income.[6] Third, we have assumed that differentiation can perfectly distinguish business from personal travel. For this reason, our welfare results constitute an upper bound on the gain from any real differentiated regime that would inevitably mis-classify some trips. Finally, we have assumed that all travel is either purely business or purely personal, abstracting from mixed-purpose trips; we extend our model to consider this possibility in the next section.

6 Extensions

This section considers three extensions. We characterize the optimal common deduction rate when cross-price responses are nonzero, when the deduction rate is constrained to be uniform across taxpayers, and when there are multiple indistinguishable goods that vary in their business versus consumption nature.

6.1 Allowing Cross-Price Effects Between the Pooled Goods

The baseline analysis assumes that compensated demand for each pooled good is unaffected by the price of the other. This assumption allows the response to a change in the common deduction rate to be summarized by the own-price spending-response terms $S_d(x)$ and $S_e(x)$. In many applications, however, the consumption good and the business input may be substitutes or complements. A change in the common deduction rate then affects spending on each good both through its own after-tax price and through the after-tax price of the other good.

We now relax the no-cross-price-effects assumption. In addition to the compensated own-price

[6] In other words, the empirical estimates may embed income effects, labor-supply changes, and substitution across routes and destinations that do not correspond to the experiment in the model. That being said, income effects are likely to be small in this setting, since typical income elasticities for airfares are near unity while the budget share of airfare in total consumer spending is low.

elasticities defined above, define the signed compensated cross-price elasticities by

$$\varepsilon^c_{de}(x) \equiv \frac{\partial d^c(x)}{\partial P_e}\frac{P_e}{d(x)}, \qquad \varepsilon^c_{ed}(x) \equiv \frac{\partial e^c(x)}{\partial P_d}\frac{P_d}{e(x)}. \tag{23}$$

Thus, a positive cross-price elasticity denotes compensated substitutability, while a negative cross-price elasticity denotes compensated complementarity.

Because a change in the common deduction rate moves the after-tax prices of both goods proportionally, the relevant behavioral object is the net response of spending on each good to this common price change. Define

$$\widetilde{S}_d(x) \equiv \big(\varepsilon^c_d(x) - \varepsilon^c_{de}(x)\big)p_d d(x), \tag{24}$$

$$\widetilde{S}_e(x) \equiv \big(\varepsilon^c_e(x) - \varepsilon^c_{ed}(x)\big)p_e e(x). \tag{25}$$

Note that Hicksian symmetry implies

$$\varepsilon^c_{de}(x)p_d d(x) = \varepsilon^c_{ed}(x)p_e e(x). \tag{26}$$

Thus, the expenditure-weighted cross-price effect is the same on the two margins. When the goods are substitutes, this common cross-effect reduces both $\widetilde{S}_d(x)$ and $\widetilde{S}_e(x)$ relative to their own-price counterparts $S_d(x)$ and $S_e(x)$. When the goods are complements, it increases both.

These terms generalize the spending-response measures in (14). For example, when the goods are substitutes, a reduction in the after-tax price of d raises compensated demand for d, while the simultaneous reduction in the price of e pushes demand for d in the opposite direction. The net response is therefore the own-price effect less the cross-price effect. When the goods are complements, the cross-price elasticity is negative, so the two effects reinforce one another. If cross-price effects are absent, then $\widetilde{S}_i(x) = S_i(x)$ for $i \in \{d, e\}$.

The optimal common deduction retains the same form as in Proposition 1, with the own-price spending responses replaced by these net responses; the proof can be found in Appendix C.

Proposition 3 (Optimal Deduction with Cross-Price Effects)**.** *Consider the tax function given*

in Proposition 1, but allow compensated cross-price effects between d and e. Any interior Pareto optimal common deduction rate at labor income level x satisfies

$$\alpha(x) = \frac{\widetilde{S}_e(x)}{\widetilde{S}_e(x) + \big(1 - T'(z(x))\big)\widetilde{S}_d(x)}. \tag{27}$$

Intuitively, once cross-price effects are allowed, the relevant behavioral response for each good is its net response to the common movement in both after-tax prices induced by a change in the deduction rate. When the goods are substitutes, these net responses are dampened relative to the own-price responses; when they are complements, they are amplified. Hicksian symmetry further implies that the same expenditure-weighted cross-price effect enters both response terms. This means that, when the goods are substitutes, the same amount is subtracted from both own-price responses. That subtraction matters proportionally more for whichever response was initially smaller, increasing the relative importance of the larger response in determining the optimal common deduction rate. Substitutability therefore pushes the rate toward the differentiated treatment of the good with the larger own-price spending response. With complementarity, the same amount is instead added to both responses, reducing their relative difference and pushing the optimal deduction in the opposite direction. Thus, substitutability makes differences in spending responsiveness more important for the optimal common deduction rate, whereas complementarity makes the two margins behave more similarly and reduces the importance of those differences.

6.2 Uniform Deduction Rate

Our main analysis assumes that the deduction rate can vary with labor income. We now impose the additional restriction that the common deduction rate must be uniform across taxpayers at different labor-income levels. The tax system therefore takes the form

$$T\left(y(x,e) - \alpha(p_d d + p_e e) + \gamma(x)\right), \tag{28}$$

where the scalar α applies to all taxpayers.

Under the income-dependent policy considered in Proposition 1, the government can perturb

the deduction rate separately at each labor-income level. In contrast, under the uniform deduction restriction considered here, the government can change α only uniformly. The relevant optimality condition consequently balances the compensated revenue effects of the deduction across the entire distribution of labor income.

For notational convenience, define the tax-wedge factor

$$\omega(x;\alpha) \equiv \frac{\big(T'(z(x))\big)^2}{\big(1-T'(z(x))\big)\big(1-\alpha T'(z(x))\big)}. \tag{29}$$

The following proposition characterizes the Pareto optimal income-invariant common deduction, with the proof located in Appendix D.

Proposition 4 (Optimal Uniform Deduction Rate)**.** *Consider tax functions of the form*

$$T\left(y(x,e) - \alpha(p_d d + p_e e) + \gamma(x)\right),$$

where the common deduction rate α is constant across labor-income levels. Absent cross-price substitution effects for d and e, any interior Pareto optimal deduction rate satisfies

$$\alpha = \frac{\displaystyle\int \omega(x;\alpha)\, S_e(x)\, \mathrm{d}H(x)}{\displaystyle\int \omega(x;\alpha)\left[S_e(x) + \big(1-T'(z(x))\big)S_d(x)\right]\, \mathrm{d}H(x)}, \tag{30}$$

where $S_d(x)$ and $S_e(x)$ are the compensated spending-response terms defined in (14) *and $\omega(x;\alpha)$ is given by* (29)*.*

Proposition 4 is the aggregate counterpart to Proposition 1. Requiring a single deduction rate prevents the government from balancing the business and consumption margins separately at each income level. Instead, the optimal uniform rate takes the form of a weighted average of the income-specific deduction rates that would be optimal under Proposition 1. Greater weight is placed on income levels where changes in deductibility generate larger compensated spending responses and larger fiscal consequences, so that the choice of deduction rate matters more for welfare and revenue.

6.3 Multiple Mixed-Purpose Goods

The baseline model separates deductible expenditures into a pure consumption good and a pure business input. In many applications, however, the expenditures included in a pooled category may span a range of goods that differ in the extent to which they provide personal consumption benefits or contribute to income production. A meal, vehicle, computer, or trip may serve a predominantly personal purpose, a predominantly business purpose, or some combination of the two. We therefore extend the model to allow for an arbitrary finite number of mixed-purpose goods.

Let $J \geq 1$ denote the number of goods in the pooled category, indexed by $j \in \{1, \ldots, J\}$, with prices p_j. At labor income level x, let $a_j(x)$ denote the quantity of good j, let

$$\mathbf{a}(x) \equiv \big(a_1(x), \ldots, a_J(x)\big),$$

and define total pooled expenditure by

$$Q(x) \equiv \sum_{j=1}^{J} p_j a_j(x). \tag{31}$$

The taxpayer's consumption subutility and pre-tax income are now given by

$$v = v\big(c, \mathbf{a}\big), \qquad y = y\big(x, \mathbf{a}\big).$$

We maintain the regularity conditions from the baseline model and focus on interior solutions. The government cannot condition deductibility on the identity of the good and instead applies the common deduction rate $\alpha(x)$ to total pooled expenditure. Taxable income is therefore

$$z(x) = y\big(x, \mathbf{a}(x)\big) - \alpha(x)Q(x) + \gamma(x). \tag{32}$$

Fixing labor income x, the taxpayer chooses c and $\mathbf{a}$ to maximize consumption subutility subject to

$$c + Q = y\big(x, \mathbf{a}\big) - T\left(y\big(x, \mathbf{a}\big) - \alpha(x)Q + \gamma(x)\right). \tag{33}$$

For each good j, define its marginal income benefit by

$$b_j(x) \equiv \frac{\partial y\big(x, \mathbf{a}(x)\big)}{\partial a_j}, \tag{34}$$

and its direct marginal consumption benefit, expressed in units of the numeraire, by

$$r_j(x) \equiv \frac{1}{v_c(x)} \frac{\partial v\big(c(x), \mathbf{a}(x)\big)}{\partial a_j}. \tag{35}$$

At an interior solution, the taxpayer's first-order condition for good j is

$$r_j(x) = p_j\big(1 - \alpha(x)T'(z(x))\big) - \big(1 - T'(z(x))\big)b_j(x). \tag{36}$$

The first term on the right-hand side is the marginal after-tax purchase cost of the good, while the second is the after-tax value of the income it produces.

We summarize the composition of good j's marginal benefit by defining its *business-intensity index*:

$$\delta_j(x) \equiv \frac{b_j(x)}{b_j(x) + r_j(x)}. \tag{37}$$

Given non-negative income and consumption benefits, this index satisfies $\delta_j(x) \in [0, 1]$. A pure consumption good has $\delta_j(x) = 0$, a pure business input has $\delta_j(x) = 1$, and a mixed-purpose good has an interior business-intensity index.

As in the baseline model, let

$$P_j(x) \equiv p_j\big(1 - \alpha(x)T'(z(x))\big)$$

denote the after-tax price of good j, and let a_j^c denote conditional compensated demand. Define the compensated own-price elasticity

$$\varepsilon_j^c(x) \equiv -\frac{\partial a_j^c(x)}{\partial P_j} \frac{P_j(x)}{a_j(x)},$$

and, paralleling (14), define the compensated spending-response term

$$S_j(x) \equiv \varepsilon_j^c(x)\, p_j a_j(x). \tag{38}$$

For ease of exposition, we assume for this analysis that compensated cross-price effects are absent:

$$\frac{\partial a_j^c(x)}{\partial P_k} = 0 \qquad \text{for all } j \neq k. \tag{39}$$

The following result extends Proposition 1 to this multiple-good setting, with the proof in Appendix E.

Proposition 5 (Optimal Deduction with Multiple Mixed-Purpose Goods)**.** *Consider tax functions of the form*

$$T\left(y\big(x, \mathbf{a}\big) - \alpha(x)Q + \gamma(x)\right),$$

where $\alpha(x)$ denotes the common deduction rate for the J mixed-purpose goods. Absent compensated cross-price effects among the pooled goods, any interior Pareto optimal deduction rate at labor income level x satisfies

$$\alpha(x) = \frac{\displaystyle\sum_{j=1}^{J} \delta_j(x) \frac{S_j(x)}{1 - T'(z(x))\delta_j(x)}}{\displaystyle\sum_{j=1}^{J} \frac{S_j(x)}{1 - T'(z(x))\delta_j(x)}}. \tag{40}$$

Proposition 5 shows that the optimal common deduction is a weighted average of the business-intensity indices of the goods in the pool. Each good receives weight proportional to

$$\frac{S_j(x)}{1 - T'(z(x))\delta_j(x)}.$$

Thus, a good matters more for the common deduction when spending on it is more responsive to the deduction. It also matters more when its returns are more exposed to the income-tax wedge. Because spending on a more business-intensive good generates more taxable income, a given spending response has larger fiscal consequences, and this effect rises with the good's business

intensity $\delta_j(x)$ and with the marginal tax rate. Proposition 1 follows as the two-good special case with $\delta_d(x) = 0$ and $\delta_e(x) = 1$.

More generally, each good pulls the common deduction toward the rate that would be optimal for that good in isolation, $\delta_j(x)$, with greater weight placed on goods that generate larger compensated spending responses and whose returns are more exposed to the income-tax wedge. The multiple-good setting therefore preserves the same basic intuition as Proposition 1: when the government must apply one deduction rate to goods that call for different treatment, the optimal common rate balances those differentiated benchmarks according to the fiscal importance of the corresponding expenditure margins.

Finally, this extension helps clarify the connection between the current analysis and prior work on a single mixed-purpose good (Goldin et al., 2026). When $J = 1$, Proposition 5 yields $\alpha(x) = \delta_1(x)$, reproducing the result from that paper that the optimal deduction for a single mixed-purpose expenditure is determined by the business intensity of its marginal benefits. The current setting generalizes that logic to a pool containing multiple expenditures with different business intensities. With a single good, responsiveness multiplies both the fiscal gain and the fiscal cost of a change in the deduction rate, so it scales the whole optimality condition without shifting where it is satisfied. The optimum is then simply the good's own business intensity. With several goods, by contrast, each good's responsiveness multiplies only its own margin in the optimality condition, so the more responsive goods account for more of the fiscal consequence and the optimal rate settles closer to their business intensities.

7 Conclusion

In this paper, we study how tax policy should treat expenditures whose appropriate tax treatment depends on their purpose when that purpose is difficult for the tax authority to observe. When business and personal expenditures must receive a common deduction rate, the optimal policy balances the distortions on the two margins according to their compensated spending responses and the income-tax wedge. More generally, when a pooled category contains goods with different mixtures of business and consumption benefits, the optimal deduction is a weighted average of the treat-

ment appropriate for the individual goods, where more weight falls on goods whose compensated spending is more elastic and whose returns are more exposed to the income-tax wedge.

These results also help identify when greater differentiation in the tax code is worth its administrative and compliance costs. The welfare cost of pooling is largest when expenditures on both business and personal uses respond substantially to deductibility and when marginal income-tax rates are high. When one margin is relatively unresponsive, by contrast, a simple pooled rule can approximate differentiated treatment at relatively low efficiency cost. The case for more fact-intensive tax rules therefore depends not only on how different the underlying uses are, but also on how strongly taxpayers respond to the tax distinction.

Declaration of generative AI use.

Generative AI was used to aid in the analysis and writing of this work. The authors take full responsibility for the contents of the published article.

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A Proof of Proposition 1

The proof proceeds by first characterizing the optimality condition in terms of the response to a marginal deduction reform, then expressing this response through Hicksian demand derivatives, and finally obtaining the desired Hicksian-based optimality condition.

Step 1: We derive the optimality condition based on the marginal demand effects of a deduction reform.

Consider a reform that perturbs the deduction rate for individuals with labor income x by a small amount $\Delta\alpha(x)$. For each x and each sufficiently small $\Delta\alpha(x)$, choose an accompanying adjustment $\Delta\gamma\big(x;\Delta\alpha(x)\big)$ that exactly preserves fixed-x indirect subutility:

$$v^*\left(x;T,\alpha(x)+\Delta\alpha(x),\gamma(x)+\Delta\gamma\big(x;\Delta\alpha(x)\big)\right)=v^*\left(x;T,\alpha(x),\gamma(x)\right). \tag{41}$$

Normalize the compensating adjustment so that

$$\Delta\gamma(x;0)=0.$$

Differentiating (41) with respect to $\Delta\alpha(x)$ at $\Delta\alpha(x)=0$ and applying the envelope theorem gives

$$0=v_cT'(z(x))\left[p_dd(x)+p_ee(x)-\left.\frac{\mathrm{d}\Delta\gamma\big(x;\Delta\alpha(x)\big)}{\mathrm{d}\Delta\alpha(x)}\right|_{\Delta\alpha=0}\right].$$

Because $v_c>0$ and $T'(z(x))>0$, the compensating path satisfies

$$\left.\frac{\mathrm{d}\Delta\gamma\big(x;\Delta\alpha(x)\big)}{\mathrm{d}\Delta\alpha(x)}\right|_{\Delta\alpha=0}=p_dd(x)+p_ee(x). \tag{42}$$

By construction, fixed-x indirect subutility is unchanged along the compensating path for every labor income level x. Therefore, the utility associated with each possible value of labor supply is the same before and after the combined reform. The labor supply that maximized U^* before the reform consequently continues to maximize U^* after the reform. Thus, labor supply and the distribution of labor income are unchanged along the reform path, and the reform affects government revenue only through the behavioral responses of d and e.

We can now derive the first-order necessary condition for Pareto optimality under the common deduction $\alpha(x)$ using a reform along these lines. We start from a baseline policy (T,α,γ) and consider a small perturbation $\Delta\alpha(x)$, paired with the exact compensating adjustment $\Delta\gamma\big(x;\Delta\alpha(x)\big)$ defined by (41). Because the compensating adjustment is pinned down by $\Delta\alpha(x)$, we simplify

notation by indexing the reform in terms of $\Delta\alpha(x)$ alone. Under the reform, choices adjust to $(\hat{d}(x;\Delta\alpha), \hat{e}(x;\Delta\alpha))$, and taxable income becomes

$$\begin{aligned}\hat{z}\big(x;\Delta\alpha(x)\big) = y(x,\hat{e}(x;\Delta\alpha(x))) - \big(\alpha(x)+\Delta\alpha(x)\big)\Big(p_d\hat{d}(x;\Delta\alpha(x)) + p_e\hat{e}(x;\Delta\alpha(x))\Big) \\ + \gamma(x) + \Delta\gamma\big(x;\Delta\alpha(x)\big).\end{aligned} \tag{43}$$

Let H denote the distribution of labor income x (which is unchanged under the subutility-preserving reform). Total revenue following the reform is thus given by:

$$R(\Delta\alpha) = \int T\big(\hat{z}\big(x;\Delta\alpha(x)\big)\big)\ \mathrm{d}H(x). \tag{44}$$

Pareto optimality of the baseline policy implies that $\Delta\alpha \equiv 0$ locally maximizes (44); otherwise, a reform could leave everyone equally well off but generate additional revenue, which, if rebated lump-sum, would yield a strict welfare gain (see Kaplow (2006)).

Note that the objective in (44) is additively separable across x: the contribution of income level x depends on $\Delta\alpha(x)$ but not on the derivative $\Delta\alpha'(x)$. Therefore, the Euler–Lagrange condition for maximizing (44) reduces to a *pointwise* first-order condition: for each x,

$$\frac{\mathrm{d}}{\mathrm{d}\,\Delta\alpha(x)} T\big(\hat{z}\big(x;\Delta\alpha(x)\big)\big)\Big|_{\Delta\alpha=0} = 0 \tag{45}$$

where the derivative captures both the partial effect of the change to $\alpha(x)$ as well as the change to $\gamma(x)$ that accompanies it as part of the reform. Because our objective is to characterize when a baseline policy maximizes (44), the derivative in (45) is evaluated at $\Delta\alpha = 0$.

To understand the intuition underlying (45), suppose that the baseline policy failed to satisfy the first-order condition at some particular income level x_0, so that a reform involving a small perturbation $\Delta\alpha(x_0)$ would raise revenue. Then the government could implement a revenue-increasing reform by changing the schedule only in a small neighborhood of x_0, leaving $\alpha(x)$ and $\gamma(x)$ unchanged elsewhere. Because the reform is compensated pointwise, it leaves taxpayers' utilities unchanged, yet it raises revenue—contradicting Pareto optimality. Multiplying by the (strictly positive) density $h(x) = H'(x)$ does not change the condition.

To characterize (45), apply the chain rule to obtain

$$0 = \frac{\mathrm{d}}{\mathrm{d}\Delta\alpha(x)} T(\hat{z}(x;\Delta\alpha(x)))\Big|_{\Delta\alpha=0} = T'(z(x))\ \frac{\mathrm{d}\hat{z}(x;\Delta\alpha(x))}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0}. \tag{46}$$

In words, the reform affects revenue through its effect on taxable income, described by $\frac{d\hat{z}(x;\Delta\alpha(x))}{d\Delta\alpha(x)}\Big|_{\Delta\alpha=0}$. To characterize this term, we differentiate (43) with respect to $\Delta\alpha(x)$ and then evaluate at $\Delta\alpha = 0$. Using the identities $\hat{d}(x;0) = d(x)$ and $\hat{e}(x;0) = e(x)$, we obtain:

$$\begin{aligned}\frac{\mathrm{d}\hat{z}(x;\Delta\alpha(x))}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0} &= y_e(x,e(x))\cdot e_\alpha(x) - \Big(p_d d(x) + p_e e(x)\Big) \\ &\quad - \alpha(x)\Big(p_d d_\alpha(x) + p_e e_\alpha(x)\Big) + \frac{\mathrm{d}\Delta\gamma\big(x;\Delta\alpha(x)\big)}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0}\end{aligned} \tag{47}$$

where, for notational convenience, we have defined

$$d_\alpha(x) \equiv \frac{\mathrm{d}\hat{d}(x;\Delta\alpha(x))}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0}, \qquad e_\alpha(x) \equiv \frac{\mathrm{d}\hat{e}(x;\Delta\alpha(x))}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0}.$$

Note that d_α and e_α are *total* derivatives along the reform's path, i.e. they incorporate the induced change in γ via (42).

By the derivative of the exact compensation rule in (42), the mechanical terms in (47) cancel exactly:

$$-\big(p_d d(x) + p_e e(x)\big) + \frac{\mathrm{d}\Delta\gamma\big(x;\Delta\alpha(x)\big)}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0} = 0. \tag{48}$$

Substituting (48) into (47) yields the purely behavioral expression:

$$\frac{\mathrm{d}\hat{z}(x;\Delta\alpha(x))}{\mathrm{d}\Delta\alpha(x)}\Big|_{\Delta\alpha=0} = \big(y_e(x,e(x)) - p_e\alpha(x)\big)\, e_\alpha(x) - p_d\alpha(x)\, d_\alpha(x). \tag{49}$$

Substituting (49) into (45) gives the pointwise necessary condition for Pareto optimality:

$$T'(z(x))\Big[\big(y_e(x,e(x)) - p_e\alpha(x)\big)\, e_\alpha(x) - p_d\alpha(x)\, d_\alpha(x)\Big] = 0 \qquad \text{for all } x. \tag{50}$$

That is, the reform must not induce taxpayers to either increase or decrease taxable income.

Note that the individual first-order condition for e, given by (12), can be rearranged as:

$$y_e - p_e\alpha = \frac{p_e(1-\alpha)}{1-T'}. \tag{51}$$

Substituting (51) into (50) and rearranging yields the reform-based optimality condition

$$\alpha(x) = \frac{\frac{p_e e_\alpha(x)}{1-T'(z(x))}}{\frac{p_e e_\alpha(x)}{1-T'(z(x))} + p_d d_\alpha(x)}. \tag{52}$$

Step 2: We convert the marginal demand effects of the deduction perturbation into marginal Hicksian demands.

Because the compensating adjustment to γ holds subutilities fixed along the reform path, the responses d_α and e_α used in (52) are already compensated. In particular, they contain no income effect. What remains is to express them through standard Hicksian derivatives. We do so by showing that the compensated deduction reform perturbs the taxpayer's problem identically to a conditional Hicksian experiment that lowers the after-tax prices of d and e, so the two induce the same demand responses.

Fix a labor income level x and suppress x arguments where doing so does not create ambiguity. We use the expenditure-minimization problem dual to the taxpayer's fixed-x subutility-maximization problem. This allows us to compare the reform directly to a standard Hicksian price experiment.

For any bundle (c, d, e), define the amount of untaxed lump-sum resources required to finance that bundle under the baseline policy by

$$\mathcal{E}(c, d, e; \alpha, \gamma) = c + p_d d + p_e e - y(x, e) + T(y(x, e) - \alpha(p_d d + p_e e) + \gamma)\,. \tag{53}$$

Thus, if an untaxed transfer m is added to the taxpayer's resources, the bundle is feasible when

$$m = \mathcal{E}(c, d, e; \alpha, \gamma).$$

The Hicksian demands at the baseline subutility level solve

$$\min_{c,d,e} \ \mathcal{E}(c,d,e;\alpha,\gamma) \qquad \text{subject to} \qquad v(c,d) \geq v^*. \tag{54}$$

This is the compensated expenditure problem associated with the taxpayer's fixed-x choice problem: the target consumption subutility is held fixed, and the minimized value of $\mathcal{E}$ is the amount of outside wealth required to attain it. The resulting demands are therefore conditional Hicksian demands. Because the policy reform preserves fixed-x indirect subutility for every possible x, the outer labor-supply problem is unchanged and the taxpayer optimally selects the same labor supply.

Let λ denote the multiplier on the subutility constraint. At an interior solution, the first-order conditions for (54) are

$$\mathcal{E}_c = \lambda v_c, \tag{55}$$

$$\mathcal{E}_d = \lambda v_d, \tag{56}$$

$$\mathcal{E}_e = 0, \tag{57}$$

$$v(c,d) = v^*. \tag{58}$$

At the baseline policy (for an arbitrary bundle), the partial derivatives of $\mathcal{E}$ have a natural interpretation as marginal after-tax costs. Because an additional unit of c costs one dollar and does not affect taxable income,

$$\mathcal{E}_c = 1. \tag{59}$$

An additional unit of d costs p_d dollars, but reduces taxable income by αp_d and therefore reduces tax liability by $\alpha p_d T'(z)$. Hence,

$$\mathcal{E}_d = p_d\big(1 - \alpha T'(z)\big). \tag{60}$$

Finally, an additional unit of e costs p_e dollars, raises pre-tax income by y_e, and raises taxable income by $y_e - \alpha p_e$. Its net effect on required outside resources is therefore

$$\mathcal{E}_e = p_e - y_e + T'(z)(y_e - \alpha p_e) = p_e\big(1 - \alpha T'(z)\big) - y_e\big(1 - T'(z)\big). \tag{61}$$

Dividing the first-order condition (56) by (55) and using the partial derivatives (59) and (60) gives

$$v_d = \big(1 - \alpha T'(z)\big) p_d v_c.$$

This condition yields (11), the taxpayer's first-order condition for d in the utility maximization problem. Similarly, combining (57) and (61) gives

$$p_e\big(1 - \alpha T'(z)\big) = y_e\big(1 - T'(z)\big).$$

This condition equates the after-tax marginal cost of e to its after-tax marginal return, as stated in (12), the taxpayer's first-order condition for e in the utility maximization problem.

We now determine how the joint policy reform perturbs this expenditure-minimization problem. Under the reform, the required amount of outside wealth becomes

$$\begin{aligned}\mathcal{E}^R(c, d, e; \Delta\alpha) = c + p_d d + p_e e - y(x, e) \\ + T\big(y(x, e) - \big(\alpha + \Delta\alpha\big)(p_d d + p_e e) + \gamma + \Delta\gamma(\Delta\alpha)\big)\,,\end{aligned} \tag{62}$$

where

$$\left.\frac{\mathrm{d}\Delta\gamma(\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0} = p_d d + p_e e \tag{63}$$

and the quantities on the right-hand side are the baseline choices.

Because the reform holds subutility fixed, the post-reform choices can equivalently be characterized as the solution to the expenditure-minimization problem with objective (62) and target subutility v^*. We therefore compare the direct effect of $\Delta\alpha$ on the first-order conditions of that problem with the direct effect of a compensated Hicksian price change.

Begin with the first-order condition for d. Under the reform,

$$\mathcal{E}_d^R = p_d - \big(\alpha + \Delta\alpha\big) p_d T'\big(z^R\big)\,, \tag{64}$$

where

$$z^R = y(x,e) - \big(\alpha + \Delta\alpha\big)(p_d d + p_e e) + \gamma + \Delta\gamma(\Delta\alpha).$$

To calculate the direct perturbation of the first-order condition, hold (c,d,e) fixed and differentiate (64) with respect to $\Delta\alpha$. Evaluating at the baseline gives

$$\begin{aligned}\left.\frac{\partial \mathcal{E}_d^R}{\partial \Delta\alpha}\right|_{\substack{\Delta\alpha=0\\(c,d,e)\ \text{fixed}}} &= -\,p_d T'(z) - \alpha p_d T''(z)\left[-(p_d d + p_e e) + \left.\frac{\mathrm{d}\Delta\gamma}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0}\right]\\ &= -\,p_d T'(z), \end{aligned} \tag{65}$$

where the second equality follows from (63). Thus, the terms involving the curvature of the tax schedule cancel from the direct perturbation of the d first-order condition.

The same calculation applies to the first-order condition for e. Under the reform,

$$\mathcal{E}_e^R = p_e - y_e + T'\big(z^R\big)\,\big(y_e - (\alpha + \Delta\alpha)p_e\big).$$

Holding (c,d,e) fixed and differentiating with respect to $\Delta\alpha$ gives

$$\begin{aligned}\left.\frac{\partial \mathcal{E}_e^R}{\partial \Delta\alpha}\right|_{\substack{\Delta\alpha=0\\(c,d,e)\ \text{fixed}}} &= T''(z)\left[-(p_d d + p_e e) + \left.\frac{\mathrm{d}\Delta\gamma}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0}\right]\big(y_e - \alpha p_e\big) - p_e T'(z)\\ &= -\,p_e T'(z). \end{aligned} \tag{66}$$

Again, the term involving $T''(z)$ vanishes by the compensation rule. The direct perturbation of the first-order condition for c is zero, since

$$\mathcal{E}_c^R = 1. \tag{67}$$

Finally, the subutility constraint (58) has no direct dependence on the policy instruments.

Equations (65) and (66) show that the reform directly lowers the marginal after-tax purchase-price component of d at the rate $p_d T'(z)$ and that of e at the rate $p_e T'(z)$ per unit increase in α. All other direct terms in the first-order-condition system are unchanged.

Now consider a conditional Hicksian price experiment within the fixed-x expenditure problem

that directly changes the after-tax prices of d and e. Let $P_d = \big(1-\alpha T'(z)\big)p_d$ and $P_e = \big(1-\alpha T'(z)\big)p_e$ denote those prices, and perturb the expenditure objective according to

$$\mathcal{E}^H(c,d,e;\Delta\alpha) = \mathcal{E}(c,d,e;\alpha,\gamma) - T'(z)\Delta\alpha\, p_d d - T'(z)\Delta\alpha\, p_e e. \tag{68}$$

The target subutility remains v^*. Hence, minimizing (68) subject to the subutility target is a standard Hicksian experiment in which

$$\mathrm{d}P_d = -\,p_d T'(z)\,\mathrm{d}\Delta\alpha, \qquad \mathrm{d}P_e = -\,p_e T'(z)\,\mathrm{d}\Delta\alpha. \tag{69}$$

The minimized value of (68) adjusts endogenously to keep subutility fixed. By the envelope theorem, its derivative with respect to $\Delta\alpha$ at the baseline $\Delta\alpha = 0$ is

$$-\,T'(z)(p_d d + p_e e),$$

which is the usual compensating change in outside wealth associated with the price reduction.

The direct derivative of the d first-order condition under (68) is $-p_d T'(z)$; the direct derivative of the e first-order condition is $-p_e T'(z)$; and the direct derivative of the c first-order condition is zero. These are exactly the direct perturbations in (65), (66), and (67). Moreover, the direct derivative of the subutility constraint is likewise zero. The policy reform and the Hicksian price experiment also share the same baseline first-order-condition system. Consequently, the implicit function theorem implies that the two experiments induce the same derivatives of the taxpayer's choices. It follows that:

$$d_\alpha = -\,T'(z)\left(p_d\frac{\partial d^c}{\partial P_d} + p_e\frac{\partial d^c}{\partial P_e}\right), \tag{70}$$

$$e_\alpha = -\,T'(z)\left(p_d\frac{\partial e^c}{\partial P_d} + p_e\frac{\partial e^c}{\partial P_e}\right). \tag{71}$$

The maintained absence of cross-price substitution effects implies

$$\frac{\partial d^c}{\partial P_e} = 0, \qquad \frac{\partial e^c}{\partial P_d} = 0.$$

Therefore,

$$d_\alpha = -\,p_d T'(z)\frac{\partial d^c}{\partial P_d}, \qquad e_\alpha = -\,p_e T'(z)\frac{\partial e^c}{\partial P_e}. \tag{72}$$

Step 3: We substitute the marginal demand effects to arrive at the optimality condition in terms of Hicksian own-price elasticities.

Using the definition of the own-price elasticities in (13), we have

$$\varepsilon_d^c = -\,\frac{\partial d^c}{\partial P_d}\frac{P_d}{d}, \qquad \varepsilon_e^c = -\,\frac{\partial e^c}{\partial P_e}\frac{P_e}{e}. \tag{73}$$

Note that these elasticities can equivalently be expressed in pre-tax price units under a suitably defined compensated price experiment. In particular, suppose that a hypothetical change in p_j is accompanied by an adjustment to γ that holds taxable income fixed at the baseline bundle, while the minimized expenditure adjusts endogenously to preserve fixed-x subutility. Holding taxable income fixed also holds $T'(z)$ fixed, so that

$$\mathrm{d}P_j = \big(1 - \alpha T'(z)\big)\,\mathrm{d}p_j.$$

Because

$$P_j = \big(1 - \alpha T'(z)\big)p_j,$$

the proportional changes in P_j and p_j are identical under this tax-base-neutralized compensated experiment. The elasticity is therefore invariant to whether it is expressed in P_j units or in p_j units for that experiment.[7]

[7]This equivalence does not apply to an unaccompanied change in the actual market price inside the nonlinear tax base, which may also change $T'(z)$.

Using $P_d = \big(1 - \alpha T'(z)\big)p_d$ and $P_e = \big(1 - \alpha T'(z)\big)p_e$ in (73) gives

$$\frac{\partial d^c}{\partial P_d} = -\frac{\varepsilon_d^c d}{\big(1 - \alpha T'(z)\big)p_d}, \qquad \frac{\partial e^c}{\partial P_e} = -\frac{\varepsilon_e^c e}{\big(1 - \alpha T'(z)\big)p_e}.$$

Substituting these expressions into (72) yields

$$d_\alpha = \frac{T'(z)}{1 - \alpha T'(z)}\varepsilon_d^c d, \qquad e_\alpha = \frac{T'(z)}{1 - \alpha T'(z)}\varepsilon_e^c e. \tag{74}$$

Finally, substituting (74) into the reform-based optimality condition (52) and canceling the common factor

$$\frac{T'(z)}{1 - \alpha T'(z)}$$

gives

$$\begin{aligned}\alpha(x) &= \frac{\frac{\varepsilon_e^c(x)p_e e(x)}{1 - T'(z(x))}}{\frac{\varepsilon_e^c(x)p_e e(x)}{1 - T'(z(x))} + \varepsilon_d^c(x)p_d d(x)} \\ &= \frac{\varepsilon_e^c(x)p_e e(x)}{\varepsilon_e^c(x)p_e e(x) + \big(1 - T'(z(x))\big)\varepsilon_d^c(x)p_d d(x)}.\end{aligned}$$

This completes the proof. □

B Proof of Proposition 2

Fix a labor income level x and suppress the x argument where doing so does not create ambiguity. We begin at the optimal pooled policy, so that

$$\alpha_d = \alpha_e = \alpha,$$

and consider compensated reforms that move the two deduction rates to the differentiated values

$$(\alpha_d, \alpha_e) = (0, 1).$$

Under differentiated deduction rates, taxable income is

$$z = y(x, e) - \alpha_d p_d d - \alpha_e p_e e + \gamma. \tag{75}$$

Consider a small reform that changes the deduction rates by $\Delta\alpha_d$ and $\Delta\alpha_e$, induces behavioral changes Δd and Δe, and changes the auxiliary tax-base term by $\Delta\gamma$. By the envelope theorem, the reform preserves subutility to first order when

$$\Delta\gamma = p_d d \Delta\alpha_d + p_e e \Delta\alpha_e. \tag{76}$$

As in the proof of Proposition 1, preserving subutility at every labor income level also preserves labor supply. The distribution H of labor income is therefore unchanged along the reform path.

To obtain the change in taxable income, subtract the baseline tax base (75) from the post-reform tax base and retain first-order terms. This gives

$$\begin{aligned} \Delta z &\approx y_e \Delta e - p_d d \Delta\alpha_d - \alpha_d p_d \Delta d - p_e e \Delta\alpha_e - \alpha_e p_e \Delta e + \Delta\gamma \\ &= \big(y_e - \alpha_e p_e\big)\Delta e - \alpha_d p_d \Delta d, \end{aligned} \tag{77}$$

where the second equality follows from the compensation rule (76). Products of two reform-induced changes, such as $\Delta\alpha_d \Delta d$ and $\Delta\alpha_e \Delta e$, are second-order and are omitted from the first-order expression. The corresponding first-order compensated revenue change at this income level is therefore

$$\Delta R \approx T'(z)\left[\big(y_e - \alpha_e p_e\big)\Delta e - \alpha_d p_d \Delta d\right]. \tag{78}$$

Absent cross-price substitution effects, a compensated change in α_e affects only e, while a compensated change in α_d affects only d. Define the compensated own-policy responses at the pooled optimum by

$$e_{\alpha_e} \equiv \left.\frac{\partial e}{\partial \alpha_e}\right|_{v^*}, \qquad d_{\alpha_d} \equiv \left.\frac{\partial d}{\partial \alpha_d}\right|_{v^*}.$$

Equation (78) then implies

$$\frac{\partial R}{\partial \alpha_e} = T'(z)\big(y_e - \alpha_e\, p_e\big)e_{\alpha_e}, \tag{79}$$

$$\frac{\partial R}{\partial \alpha_d} = -T'(z)\alpha_d p_d d_{\alpha_d}. \tag{80}$$

The individual's first-order condition for the business input under differentiated deduction rates is

$$p_e\big(1 - \alpha_e T'(z)\big) = y_e\big(1 - T'(z)\big).$$

Rearranging gives

$$y_e - p_e\alpha_e = \frac{p_e\big(1-\alpha_e\big)}{1 - T'(z)}.$$

Multiplying by T' gives

$$T'(z)\big(y_e - p_e\alpha_e\big) = \frac{T'(z)}{1 - T'(z)}p_e\big(1-\alpha_e\big). \tag{81}$$

Thus, (79) implies that the marginal gain from raising α_e vanishes at $\alpha_e = 1$. Moreover, (80) implies that the marginal gain from lowering α_d vanishes at $\alpha_d = 0$. These are the differentiated benchmark rates.

We now apply the Harberger approximation. Let

$$t \equiv T'(z)$$

denote the marginal tax rate evaluated at the pooled optimum. We evaluate t, e_{α_e}, and d_{α_d} at that baseline and treat them as constant while integrating the marginal revenue effects from (α, α) to $(0, 1)$. Because cross-price effects are absent, the order in which the two deduction rates are changed is immaterial under this approximation.

First, raise α_e from α to one while holding α_d fixed. Combining (79) and (81), the approximate

gain is

$$\begin{aligned}\Delta W_e &\approx \int_{\alpha}^{1} \frac{t}{1-t} p_e (1-a) e_{\alpha_e} \, \mathrm{d}a \\ &= \frac{1}{2} \frac{t}{1-t} p_e e_{\alpha_e} (1-\alpha)^2. \end{aligned} \tag{82}$$

Second, lower α_d from α to zero while holding α_e fixed. Using (80), the approximate gain is

$$\begin{aligned}\Delta W_d &\approx \int_{0}^{\alpha} t a p_d d_{\alpha_d} \, \mathrm{d}a \\ &= \frac{1}{2} t p_d d_{\alpha_d} \alpha^2. \end{aligned} \tag{83}$$

It remains to express the policy responses in terms of the Hicksian elasticities used in Proposition 1. The argument in the proof of Proposition 1 applies separately to each own-deduction reform. At the pooled baseline, a compensated increase in either deduction rate lowers the corresponding after-tax price at rate t, while the baseline after-tax price of each good is proportional to $1 - \alpha t$. Therefore, analogously to (74) we obtain

$$e_{\alpha_e} = \frac{t}{1-\alpha t} \varepsilon_e^c e, \qquad d_{\alpha_d} = \frac{t}{1-\alpha t} \varepsilon_d^c d. \tag{84}$$

Substituting (84) into (82) and (83) gives

$$\Delta W_e \approx \frac{1}{2} \frac{t^2}{(1-t)(1-\alpha t)} \varepsilon_e^c p_e e (1-\alpha)^2, \tag{85}$$

$$\Delta W_d \approx \frac{1}{2} \frac{t^2}{1-\alpha t} \varepsilon_d^c p_d d \alpha^2. \tag{86}$$

Define

$$A \equiv \varepsilon_e^c p_e e, \qquad B \equiv (1-t) \varepsilon_d^c p_d d, \qquad \omega \equiv \frac{t^2}{(1-t)(1-\alpha t)}.$$

Adding (85) and (86) yields the local two-triangle expression

$$\Delta W(x) \approx \frac{1}{2} \omega \left[A(1-\alpha)^2 + B\alpha^2 \right]. \tag{87}$$

By Proposition 1, the optimal common deduction rate is

$$\alpha = \frac{A}{A+B}, \qquad 1-\alpha = \frac{B}{A+B}.$$

Consequently,

$$A(1-\alpha)^2 + B\alpha^2 = A\left(\frac{B}{A+B}\right)^2 + B\left(\frac{A}{A+B}\right)^2 = \frac{AB}{A+B}. \tag{88}$$

Moreover, rearranging the equation for the optimal common α, we have

$$1-\alpha t = \frac{A(1-t)+B}{A+B}. \tag{89}$$

Substituting (88) and (89) into (87) gives

$$\Delta W(x) \approx \frac{1}{2}\frac{t^2}{1-t}\frac{AB}{A(1-t)+B}. \tag{90}$$

Recalling

$$A = \varepsilon_e^c p_e e = S_e,$$
$$B = (1-t)\varepsilon_d^c p_d d = (1-t)S_d,$$

we obtain

$$\frac{AB}{A(1-t)+B} = \frac{(1-t)S_eS_d}{(1-t)\big(S_e+S_d\big)}.$$

The common factor $1-t$ cancels, leaving

$$\Delta W(x) \approx \frac{1}{2}\frac{t^2}{1-t}\frac{S_eS_d}{S_e+S_d}. \tag{91}$$

Finally, aggregating across labor income levels and substituting $t(x) = T'(z(x))$ yields

$$\Delta W \approx \frac{1}{2}\int \frac{\big(T'(z(x))\big)^2}{1-T'(z(x))}\frac{S_e(x)S_d(x)}{S_e(x)+S_d(x)}\,\mathrm{d}H(x).$$

Note that, because each reform is compensated, this revenue gain is a money-metric welfare gain: it is the additional revenue that can be redistributed lump-sum while preserving taxpayers' utilities. This completes the proof. □

C Proof of Proposition 3

Fix a labor income level x and suppress the x argument where doing so does not create ambiguity. The compensation rule is unchanged from the proof of Proposition 1:

$$\Delta\gamma = (p_d d + p_e e)\Delta\alpha.$$

Consequently, the mechanical effects of the reform on taxable income cancel, and the pointwise first-order condition for an interior Pareto optimum remains

$$T'(z)\Big[(y_e - p_e\alpha)e_\alpha - p_d\alpha d_\alpha\Big] = 0, \tag{92}$$

where d_α and e_α are the compensated responses to a change in the common deduction rate. Proceeding as in the proof of Proposition 1, we express the taxpayer's first-order condition for e as

$$y_e - p_e\alpha = \frac{p_e(1-\alpha)}{1-T'(z)},$$

substitute into (92) and rearrange to obtain

$$\alpha = \frac{p_e e_\alpha}{p_e e_\alpha + \big(1-T'(z)\big)p_d d_\alpha}. \tag{93}$$

As with (52), this characterization in terms of the reform-based responses d_α and e_α does not require the absence of cross-price effects. Cross-price effects matter only for the mapping from d_α and e_α to Hicksian elasticities.

As shown in the proof of Proposition 1, the reform is locally equivalent to reducing both after-tax

prices at the rate

$$\mathrm{d}P_d = -\,p_d T'(z)\,\mathrm{d}\Delta\alpha, \qquad \mathrm{d}P_e = -\,p_e T'(z)\,\mathrm{d}\Delta\alpha.$$

The chain rule therefore gives

$$d_\alpha = -\,T'(z)\left(p_d \frac{\partial d^c}{\partial P_d} + p_e \frac{\partial d^c}{\partial P_e}\right), \tag{94}$$

$$e_\alpha = -\,T'(z)\left(p_d \frac{\partial e^c}{\partial P_d} + p_e \frac{\partial e^c}{\partial P_e}\right). \tag{95}$$

Using $P_d = \big(1-\alpha T'(z)\big)p_d$ and $P_e = \big(1-\alpha T'(z)\big)p_e$, together with the definitions of the own-price elasticities in (13) and the cross-price elasticities in (23), we obtain

$$d_\alpha = \frac{T'(z)}{1-\alpha T'(z)}\big(\varepsilon_d^c - \varepsilon_{de}^c\big)d, \tag{96}$$

$$e_\alpha = \frac{T'(z)}{1-\alpha T'(z)}\big(\varepsilon_e^c - \varepsilon_{ed}^c\big)e. \tag{97}$$

Substituting (96) and (97) into (93) and canceling the common factor $T'(z)/(1-\alpha T'(z))$ gives (27). This completes the proof. □

D Proof of Proposition 4

Consider a small uniform perturbation $\Delta\alpha$ to the common deduction rate. As in the proof of Proposition 1, accompany this perturbation with an income-dependent adjustment to $\gamma(x)$ satisfying

$$\Delta\gamma(x) = \big(p_d d(x) + p_e e(x)\big)\Delta\alpha. \tag{98}$$

This adjustment preserves fixed-x subutility at every labor-income level. It therefore also leaves labor supply, and hence the distribution H of labor income, unchanged.

Let $d_\alpha(x)$ and $e_\alpha(x)$ denote the compensated responses of the two goods to this uniform reform:

$$d_\alpha(x) \equiv \left.\frac{\mathrm{d}\hat{d}(x;\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0}, \qquad e_\alpha(x) \equiv \left.\frac{\mathrm{d}\hat{e}(x;\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0}.$$

Differentiating taxable income and using the compensation rule (98), the mechanical effects of the reform cancel. The remaining change in taxable income at labor-income level x is

$$\left.\frac{\mathrm{d}\hat{z}(x;\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0} = \big(y_e(x,e(x)) - p_e\alpha\big)e_\alpha(x) - p_d\alpha d_\alpha(x). \tag{99}$$

The compensated marginal revenue effect of the uniform reform is therefore

$$\left.\frac{\mathrm{d}R(\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0} = \int t(x)\left[\big(y_e(x,e(x)) - p_e\alpha\big)e_\alpha(x) - p_d\alpha d_\alpha(x)\right]\,\mathrm{d}H(x), \tag{100}$$

where $t(x) \equiv T'(z(x))$ denotes the marginal tax rate. At an interior optimum, (100) must equal zero.

The taxpayer's first-order condition for e implies

$$y_e(x,e(x)) - p_e\alpha = \frac{p_e(1-\alpha)}{1-t(x)}. \tag{101}$$

Moreover, the Hicksian-response argument in the proof of Proposition 1 applies separately at every labor-income level. Absent cross-price effects, we therefore have

$$d_\alpha(x) = \frac{t(x)}{1-\alpha t(x)}\varepsilon_d^c(x)d(x), \qquad e_\alpha(x) = \frac{t(x)}{1-\alpha t(x)}\varepsilon_e^c(x)e(x). \tag{102}$$

Substituting (101) and (102) into (100) gives

$$0 = \int \omega(x;\alpha)\left[(1-\alpha)\varepsilon_e^c(x)p_e e(x) - \alpha\big(1-t(x)\big)\varepsilon_d^c(x)p_d d(x)\right]\,\mathrm{d}H(x). \tag{103}$$

Equivalently,

$$\int \omega(x;\alpha)\varepsilon_e^c(x)p_e e(x)\,\mathrm{d}H(x) = \alpha\int \omega(x;\alpha)\left[\varepsilon_e^c(x)p_e e(x) + \big(1-t(x)\big)\varepsilon_d^c(x)p_d d(x)\right]\,\mathrm{d}H(x).$$

Dividing by the integral multiplying α yields (30). □

E Proof of Proposition 5

Fix a labor income level x and suppress the x argument where doing so does not create ambiguity. Recall from (31) that

$$Q = \sum_{j=1}^{J} p_j a_j$$

denotes total expenditure on the pooled goods. Consider a small perturbation $\Delta\alpha$ to the common deduction rate, accompanied by an adjustment $\Delta\gamma(\Delta\alpha)$ that preserves fixed-x subutility. As in the proof of Proposition 1, the envelope theorem implies

$$\left.\frac{\mathrm{d}\Delta\gamma(\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0} = Q. \tag{104}$$

Because the reform preserves fixed-x subutility, it also leaves labor supply unchanged.

Under the reform, choices adjust to

$$\hat{\mathbf{a}}(\Delta\alpha) = \big(\hat{a}_1(\Delta\alpha), \ldots, \hat{a}_J(\Delta\alpha)\big),$$

and taxable income becomes

$$\hat{z}(\Delta\alpha) = y\big(x, \hat{\mathbf{a}}(\Delta\alpha)\big) - (\alpha + \Delta\alpha)\sum_{j=1}^{J} p_j \hat{a}_j(\Delta\alpha) + \gamma + \Delta\gamma(\Delta\alpha). \tag{105}$$

Define the compensated policy response of good j by

$$a_{j,\alpha} \equiv \left.\frac{\mathrm{d}\hat{a}_j(\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0}. \tag{106}$$

Differentiating (105) and using (104), the mechanical effects of the deduction change and the compensating adjustment cancel. Hence,

$$\left.\frac{\mathrm{d}\hat{z}(\Delta\alpha)}{\mathrm{d}\Delta\alpha}\right|_{\Delta\alpha=0} = \sum_{j=1}^{J} \big(b_j - p_j\alpha\big) a_{j,\alpha}. \tag{107}$$

Pareto optimality requires the compensated reform to have no first-order effect on revenue. Since

$T'(z) > 0$, the pointwise first-order condition is therefore

$$\sum_{j=1}^{J} \big(b_j - p_j\alpha\big)a_{j,\alpha} = 0. \tag{108}$$

We next express the fiscal effect $b_j - p_j\alpha$ in terms of the business-intensity index δ_j. Recall from (36) that the taxpayer's first-order condition for good j is

$$r_j = p_j\big(1 - \alpha T'(z)\big) - \big(1 - T'(z)\big)b_j. \tag{109}$$

It follows that

$$b_j + r_j = p_j + T'(z)\big(b_j - p_j\alpha\big). \tag{110}$$

By the definition of the business-intensity index, we have

$$\delta_j = \frac{b_j}{b_j + r_j}.$$

Combining this definition with (110) gives

$$b_j = \delta_j\big(b_j + r_j\big) = \delta_j\big[p_j + T'(z)\big(b_j - p_j\alpha\big)\big]. \tag{111}$$

Writing $b_j = \big(b_j - p_j\alpha\big) + p_j\alpha$ on the left-hand side and solving for $b_j - p_j\alpha$ gives

$$b_j - p_j\alpha = \frac{p_j(\delta_j - \alpha)}{1 - T'(z)\delta_j}. \tag{112}$$

Substituting this expression into (108) yields

$$\sum_{j=1}^{J} \frac{p_j(\delta_j - \alpha)}{1 - T'(z)\delta_j}a_{j,\alpha} = 0. \tag{113}$$

It remains to express the compensated policy responses $a_{j,\alpha}$ in terms of the Hicksian elasticities defined in the text. Consider the expenditure-minimization problem dual to the taxpayer's fixed-x subutility-maximization problem. The amount of untaxed outside resources required to finance a

bundle is

$$\mathcal{E}(c, \mathbf{a}; \alpha, \gamma) = c + \sum_{j=1}^{J} p_j a_j - y(x, \mathbf{a}) + T\left(y(x, \mathbf{a}) - \alpha \sum_{j=1}^{J} p_j a_j + \gamma \right). \tag{114}$$

For a fixed bundle, the marginal expenditure associated with good j is

$$\mathcal{E}_{a_j} = p_j - b_j + T'(z)\big(b_j - p_j \alpha\big). \tag{115}$$

Under the compensated reform, hold the bundle fixed and differentiate this marginal condition with respect to $\Delta\alpha$, writing $\mathcal{E}^R_{a_j}$ for the resulting reform value of the marginal expenditure. The compensation rule (104) keeps taxable income fixed at the baseline bundle to first order, so the term involving $T''(z)$ vanishes. The direct shift in the marginal condition is therefore

$$\left. \frac{\partial \mathcal{E}^R_{a_j}}{\partial \Delta\alpha} \right|_{\Delta\alpha = 0} = - p_j T'(z). \tag{116}$$

Thus, the compensated deduction reform is locally equivalent to a Hicksian experiment that changes the after-tax purchase price of each pooled good according to

$$\frac{\mathrm{d}P_j}{\mathrm{d}\Delta\alpha} = - p_j T'(z). \tag{117}$$

The general compensated response is consequently

$$a_{j,\alpha} = - T'(z) \sum_{k=1}^{J} p_k \frac{\partial a_j^c}{\partial P_k}. \tag{118}$$

Absent compensated cross-price effects, this reduces to

$$a_{j,\alpha} = - p_j T'(z) \frac{\partial a_j^c}{\partial P_j}. \tag{119}$$

By definition,

$$\varepsilon_j^c = - \frac{\partial a_j^c}{\partial P_j} \frac{P_j}{a_j}, \qquad P_j = \big(1 - \alpha T'(z)\big) p_j.$$

Therefore,

$$a_{j,\alpha} = \frac{T'(z)}{1 - \alpha T'(z)} \varepsilon_j^c a_j. \tag{120}$$

Substituting (120) into (113) and canceling the common positive factor $T'(z)/(1-\alpha T'(z))$ gives

$$\sum_{j=1}^{J} \frac{(\delta_j - \alpha) S_j}{1 - T'(z)\delta_j} = 0, \tag{121}$$

where $S_j = \varepsilon_j^c p_j a_j$. Collecting the terms that multiply α gives

$$\alpha \sum_{j=1}^{J} \frac{S_j}{1 - T'(z)\delta_j} = \sum_{j=1}^{J} \delta_j \frac{S_j}{1 - T'(z)\delta_j}.$$

Dividing by the term multiplying α and restoring the x arguments yields

$$\alpha(x) = \frac{\displaystyle\sum_{j=1}^{J} \delta_j(x) \frac{S_j(x)}{1 - T'(z(x))\delta_j(x)}}{\displaystyle\sum_{j=1}^{J} \frac{S_j(x)}{1 - T'(z(x))\delta_j(x)}},$$

which is (40). This completes the proof. □